Data Structures and Algorithms · 21CSC201J

Unit 1 · Session 2 · SLO 1 — Solved Worksheet

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Unit â€“ 1 : Session â€“ 2 : SLO â€“ 1 | Define, Declare and Access a Structure

I. Fill in the Blanks

1. struct â€” a structure is defined using the struct keyword.
2. member declarations (data members) â€” struct StructName { members };
3. variableName â€” struct StructName { ... } varName;
4. dot (.) â€” members are accessed with the dot operator via a structure variable.
5. arrow (->) â€” members are accessed with the arrow operator via a pointer to a structure.
6. variableName â€” struct StructName variableName;
7. pointerName â€” struct StructName *ptr;
8. variableName â€” ptr = &variableName;
9. single (user-defined) â€” structures group different types into one user-defined type.
10. semicolon â€” structure definitions end with a semicolon.

II. Practice the Code

1. Student record: struct Student { char name[50]; int roll; float marks; }; struct Student s = {"Asha", 12, 89.5}; printf("%s %d %.1f", s.name, s.roll, s.marks);
2. Three employees: struct Emp { char name[50]; float salary; }; struct Emp e[3]; for (int i = 0; i < 3; i++) scanf("%s %f", e[i].name, &e[i].salary); // access: e[i].name, e[i].salary
3. Book with pointer: struct Book { char title[50]; float price; }; struct Book b = {"C Programming", 450}; struct Book *ptr = &b; printf("%s %.2f", ptr->title, ptr->price);